

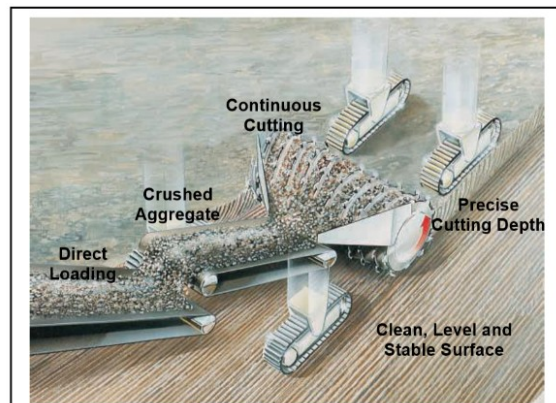
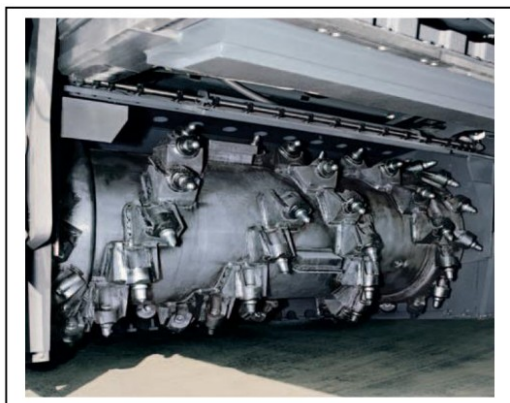
Continuous Surface Miners

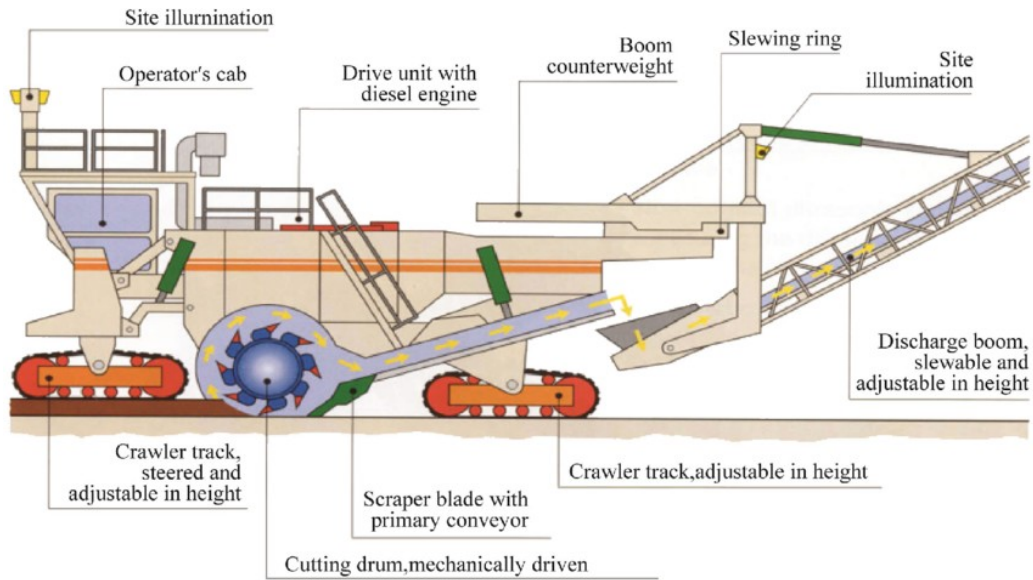
1.0 INTRODUCTION

Continuous surface miner (CSM) is basically a cutting machine used in surface mines. These machines have been developed during 1970s primarily for extraction of thin seams (not less than 2.5 cm thick) and selective removal of thin dirt bands present in seams.



The machines have single or double front crawlers and two rear crawlers. In Wirtgen Surface Miner, a type of CSM, a cutting drum, with specified number of replaceable picks mounted on it arranged in form of helix, is fitted below the machine body in between the front and rear crawlers. When the machine works, the hydraulically controlled drum rotates in up-cutting direction. The cutting drum is generally kept in horizontal position during cutting, but can be inclined to some degree by controlling the hydraulic rams to facilitate cutting of inclined seams. During its working, the cutting drum rotates in an up-cutting direction and a layer of predetermined thickness (maximum thickness depends on the machine model) of the deposit is cut and crushed in between the picks. The "Electronic Depth Controller System" fitted in the machine controls the thickness of cut slice/slices. The width of the cut slice is same as the length of the cutting drum.





1.1 Applicability

The following conditions favour surface miner:

- Selective mining
- Precise cutting of trenches, surfaces and slopes
- Ramp formation
- Multiple seams
- Material exploitation and crushing without blasting
- Haul road rehabilitation
- Inclined seams.
- Steep high wall slopes

1.2 Present Applications

1. The main fields of applications of Continuous Surface Miner, in the case of Industrial Minerals, are as follows:

- Coal
- Phosphate
- Tar sand, oil shale
- Clay
- Bauxite
- Soft Limestone
- Gypsum
- Saltpeter, etc.

Overburden and Waste Partings

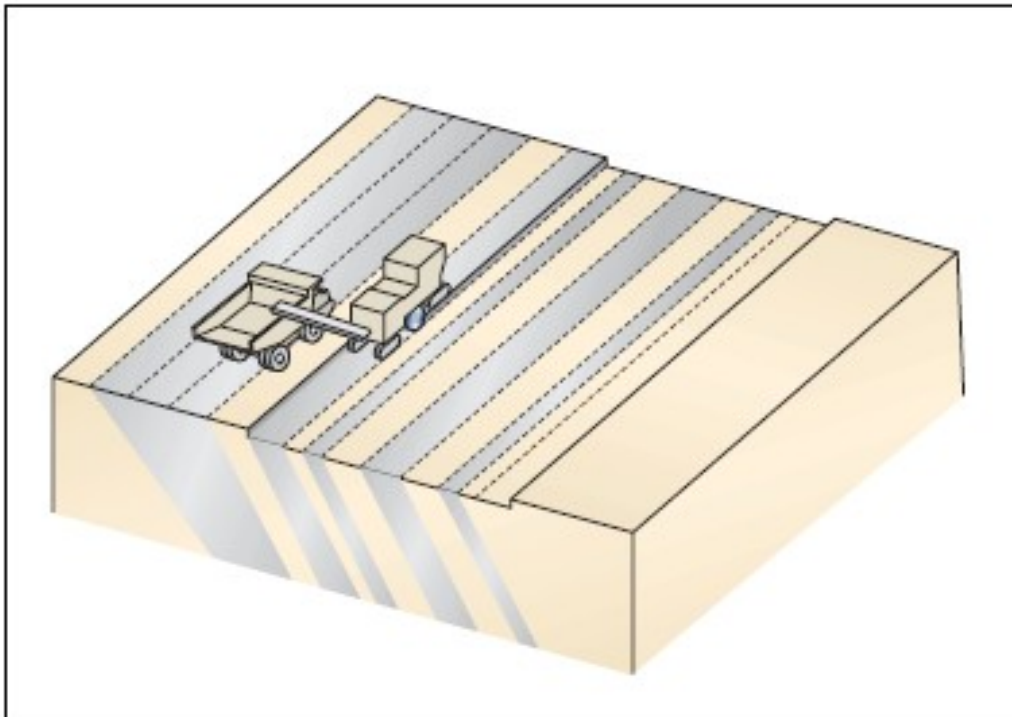
- Sand and gravel
- Mud and silt stone
- Shale
- Sandstone
- Weathered and semi-solid industrial mineral and deposits, etc

2. Blast free mining

3. In underground salt mines, these machines are used to maintain the heavily frequented transport roads.

4. Surface Miners are used in tunnels to enlarge the headroom by lowering the tunnel floor.

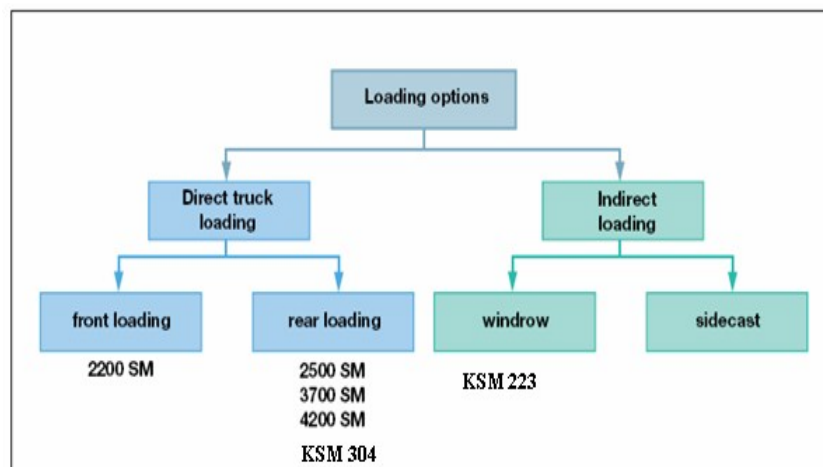
5. Cutting very steep seams or pockets of pay material can be mined as shown on the Fig. below.



Steeply inclined multiple seams

2.0 LOADING OPTIONS AND THEIR COMPARISON

The surface miner offers the loading and material handling options as shown in Fig. below and their advantages and disadvantages are given in Table below.



Classification of loading options

Table - Comparison of the different loading methods



Loading method	Advantages	Disadvantages
Direct Loading	▪ No re-handling of material required	▪ Larger working area required for truck maneuvering ▪ Production affected by truck exchange time ▪ Belt wear
Side casting	▪ Blending of material in the mine ▪ Stockpile of material in the mine ▪ No waiting for trucks, independent operation	▪ Restricted to 3-5 cuts wide on each side of the mine stockpile ▪ Belt wear ▪ Material has to be re-handled ▪ Material prone to absorb water when lying on the ground

Windrowing	▪ No waiting for trucks, ▪ No belt wear/higher availability ▪ Higher production rates than conveyor loading ▪ Coarser material ▪ Better selectivity on steep inclined seams	▪ Large working area required ▪ Material has to be handled either by loader or scraper ▪ Material prone to absorb water when lying on the ground
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Merits and demerits of surface Miner

Advantages

1. A significant advantage of this type of surface miner is that it is suitable for thin mining layers and harder minerals.
2. In contrast to BWE, this machine achieves higher separation accuracy with regard to the individual seams and strata.
3. It can be used for the hard strata or minerals, where the rock had to be blasted before the material is excavated by the crawler shovels.
4. Additional crusher systems are no longer required since the rotary grinder crushes the material up to a maximum grain size of approx. 200 mm.
5. Due to better maneuverability, this machine is suitable for the narrow sites, as it requires the little space and little time for changing the direction at the face ends.
6. This can be operated in the automatic mode with the help of the GPS control and longer service life.

Surface miner has several merits as well as demerits as given in Table below.

Table - Merits and demerits of surface miner

Merits	Demerits
i. It offers low capital cost. ii. It requires lesser infrastructure and civil work prior to machine operation. iii. Surface miner technology eliminates drilling and blasting, hence automatically the associated problems like fly rock and ground vibrations etc. are eliminated. iv. No primary crushing is required, as the size of the material produced by surface miner is below 150 mm (95%). v. As the product size is smaller and uniform the material is readily conveyable and when it is loaded on to dumper/truck, the fill factor is high. vi. It can cut layer as thin as 15 mm, which ensures selective mining of thin bands also. vii. It ensures high truck capacity utilisation because of its conveyor boom loading. viii. Shifting of the machine is also very easy as the machine is very much portable. ix. It is highly flexible within operational area. x. It produces smooth after-cut surface, which reduces the transportation cost by increasing tyre life. xi. As it has slope control facility, it can create such an inclined after cut surface, so that no extra draining operation will	i. Surface Miner productivity reduces in hard abrasive and sticky material. ii. Surface Miner cannot work in undulating ground efficiently. iii. Noise level and dust level increases if dry cutting is done and in such situations pick consumption will also be increased. iv. It is unsuitable for narrow long bench condition, as there may not be adequate turning place available for the machine. The empty travel back time may also be very high. v. It may not be economic for very thick homogeneous deposits. vi. It has to leave some place near the end of the cut (sidewall of the bench) which can be cut by cross cutting later on. But in narrow bench that length may not be extractable and most of the cases dozing is the only method to overcome the difficulty. vii. The machine is diesel powered. viii. If any mine requires larger size products with no or very less amount of fines, the machine is not suitable. ix. The performance of the machine is affected in rainy season particularly when operating in sticky material.

Merits	Demerits
be required. xii. Machine can work efficiently over a very wide temperature range (5°C to 50°C).	

3.0 PERFORMANCE ANALYSIS OF SURFACE MINERS

The applicability of surface miners has grown to even iron ore and granite deposits today and similar attempts are being made in India by expanding their application in hard limestone mines from the experience already gained. The salient observations on the performance of surface miners in global perspective are presented in Table below.

Table – A global review of surface miners performance in different geomining condition

Mine	Machine type	Rock type	Rock strength (MPa)	Production (tph)	Remarks
Marra Mamma Fortescue Chicheter	2500 SM	Iron ore	80 - 100	350 – 550	No micro-jointing present.
Eerste Nederlandsche Cement Industrie, Netherlands	2200 SM	Marl		660 – 720	Currently quarry is about 30 m below the ground water table.
OKUMURA GUMI DOBOKU, Japan	2500 SM 3500 SM	Granite	140 - 260	70 (bcum)	Trench has to be cut into a hill where houses and factories stand.
Debele mine, Guinea	2100 SM	Bauxite	30 – 125	250	
Frija mine, Guinea	2200 SM	Bauxite	30 – 125	180	Very complex deposit structure.
Frija mine, Guinea	2500 SM	Bauxite	30 - 125	360	
Devonian area, Warstein, Germany	2200 SM	Limestone	100 -125	294	Compacted with cracks spaced over 50 cm apart.
Madras cement ltd., Pennadam, India	2500 SM	Limestone	30 -50	613	

Technical Specification of a typical Surface Miner

Make: Wirtgen GmbH / Germany

Model: 2600 SM

Cutting Drum Width:

2600 mm

Cutting Drum Depth:

0 – 250 mm

Cutting Drum diameter with tools

950 mm

Number of cutting tools on the drum:

Depending upon machine application

Engine Power:

560 kW

Operating speed:

0 – 25 m/min

Ground Pressure:

2.57 bar

Gradeability

20°

Ground clearance:

300 mm

Primary conveyor width:

1000 mm

Primary conveyor length

4000 mm

Discharge conveyor width

1000 mm

Discharge conveyor length

10,000 mm

Belt speed	0 – 3 m/s
Overall length of the machine:	19270 mm
Overall height of the machine (discharge conveyor inclination)	6500 mm
Overall width of the machine	3100 mm
Slewing angle of the discharge conveyor	± 90°
Turning angle of the front crawler	± 45°
No. of crawlers	total 3 nos, (2 in rear side and one in front side)

4.0 CRITICAL ISSUES CONCERNING DEPLOYMENT OF SURFACE MINER

4.1 Location

The mining lease boundary and its surroundings (villages, rivers and nallah etc., at close proximity).

4.2 HYDROLOGICAL CONDITIONS

Depth of water table (7 to 9 Mtrs below ground level depending upon topography). The hydrological condition of the mine leads to water stagnation over benches and cannot be drained (based on topography) on its own and this affects continuous production through conventional method of mining.

4.3 QUALITY PROBLEM

This is a problem in layered strata with a variable thickness and quality.

4.4 BLASTING HAZARD

Villages which come under the danger zone of 500 Mts radius as stipulated by DGMS are the concern. The location of villages and hamlets are so close that even the costly method of controlled blasting may not be able to prevent the houses from damages due to vibration of blasting. This precludes blasting forcing to adopt mechanical cutting. With such enormous constraints, use of conventional mining methods at mines had to look for a blast free, selective mining techniques and cost effective which would also suit the unique hydrological condition prevalent at the mines. The choose was between deployment of Ripper cum dozer and surface miner both of which meet the primary requirement of blast free mining.

4.5 WHY ONE SHOULD PREFER FOR SURFACE MINER?

It is well known fact that surface miner is expensive equipment, choice, therefore was a difficult one. The first consideration was the cuttability and the resultant productivity. The formation must be highly amenable to mechanical cutting and the estimated productivity of the machine under consideration must be adequate. By proper mine planning of keeping difference in level benches surface miner can be worked smoothly and possible to feed the crusher directly with considerable dry material continuously.

Thin bands in the deposit (of low quality) can be selectively mined. The unique ability of surface miner to cut the material slice by slice in infinitely variable layer within the range would help removing of waste bands, which are easily distinguishable by color. Surface miner would thus help improving the quality of iron ore substantially.

Mining activity with surface miner would thus be carried out within close proximity of the habitats thus making available substantially additional mineable deposits, which restricts blasting to within 500 Mts of habitats.

Application of surface miners in similar restricted environment has been in practice in Madras Cements and are now the proud owners of the largest fleet of surface miners in India with four surface miners already in deployment at the limestone deposits of its two cement plants at Alathiyur and pandalgudi and the fifth one which is the largest of its kind ever used in a limestone quarry of a cement plant. This (Writgen model 2500 SM) is a 100 HP machine with over 100 tonne of operating weight, drum cutting depth up to 600 mm.

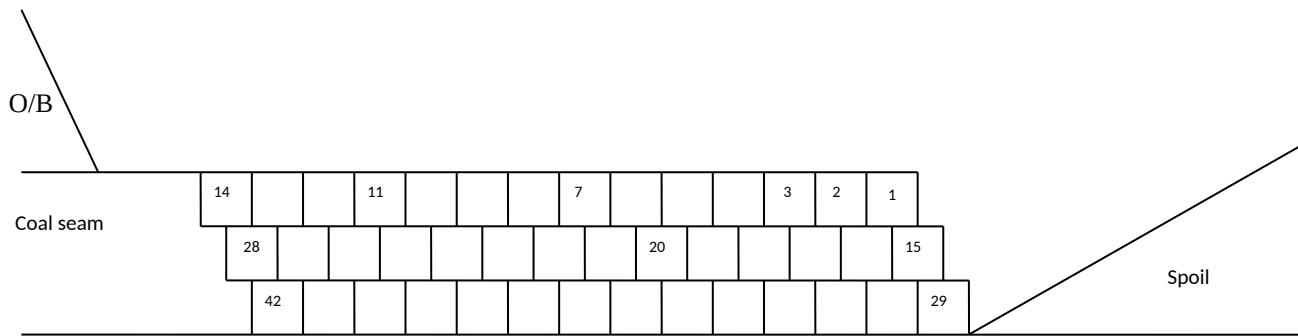
5.0 METHOD OF WORK USING CONTINUOUS SURFACE MINER

The extraction of slices may be accomplished by following depending on the situation

- a) Wide Bench Method
- b) Block Mining Method
- c) Stepped Cut Method.

5.1 Wide Bench Method

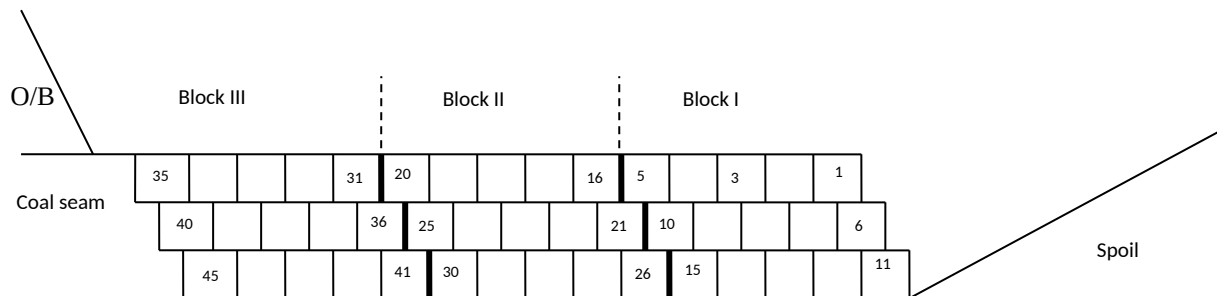
The wide bench method is generally used when the width of the mineral bench is not very high. In this method, first the uppermost layer of the whole width of the bench is extracted in number of slices followed by the extraction of the next layer and so on till the full height of the mineral bench is extracted.



Wide Bench Method

5.2 Block Mining Method

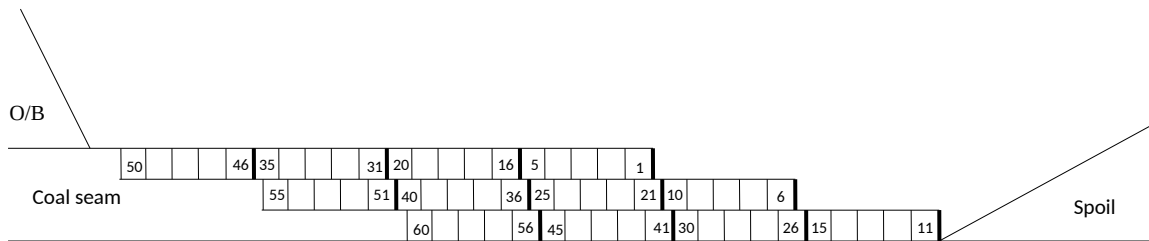
The block mining method is adopted when the mineral bench is so wide that the extraction by wide bench method is not suitable. In this method the bench width is suitably divided into number of blocks, and then the block from the edge side of the bench is extracted first for the full height of the bench in number of slices following a sequence similar to that of wide bench method. After completion of one block the next adjacent block is extracted and the process continues till the whole bench is extracted.



Block Mining Method

5.3 Stepped Cut Method

The stepped cut method is used when excess moisture/water is present in the mineral bench that creates problem in extracting the slices. In this method also the bench is first divided into suitable number of blocks and then the individual slices are so extracted that it forms a number of small steps on the mineral bench. The formation of steps facilitates the drainage from the mineral bench thereby improving the machine efficiency.



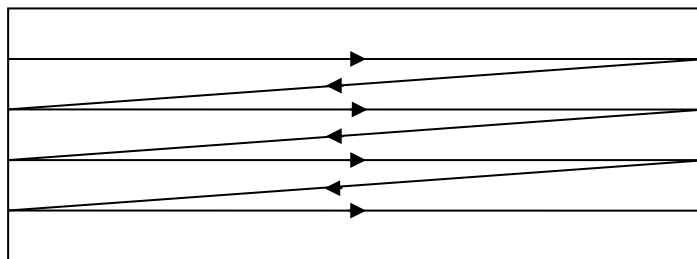
Stepped cut Method

The working method of Wirtgen Surface Miner, based on the travel mode for extraction of slices of same layer, may be classified into three categories – Empty Travel Back Method, Turn Back Method and Continuous Mining Method.

5.4 Empty Travel Back Method

In the empty travel back method, the surface miner starts excavating a slice from one end of the bench and moves upto the other end of the bench covering the full length of the bench available for extraction. After completion of one full cut, the cutting drum is raised and the machine travels back empty to the starting end. After coming back to the starting end, the machine sets for a new cut in the adjacent parallel slice, and the process continues till all the designated slices in that layer are extracted. The process is generally adopted under one or more of the following conditions.

- Available length of the bench is small (not more than 200m)
- Turning time of the machine is more than the empty travel time.
- The machine turning is difficult at the cut ends due to unavailability of required turning space.

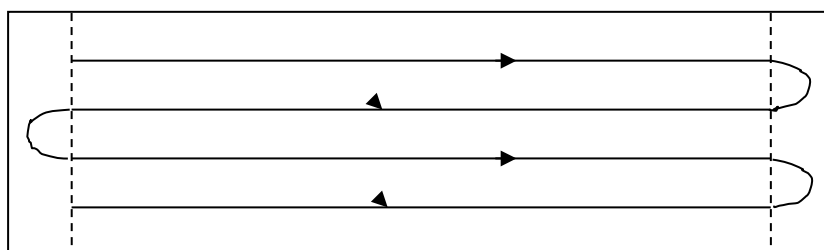


Empty Travel Back

5.5 Turn Back Method

In the turn back method, the surface miner starts cutting a slice from one end of the bench and moves to the other end covering the available length of cut. After the cut is complete the cutting drum is raised and the machine turns back there to start a fresh cut adjacent and parallel to the previous cut. In this way the required number of slices are extracted. The method is most widely accepted and used provided the following conditions exist.

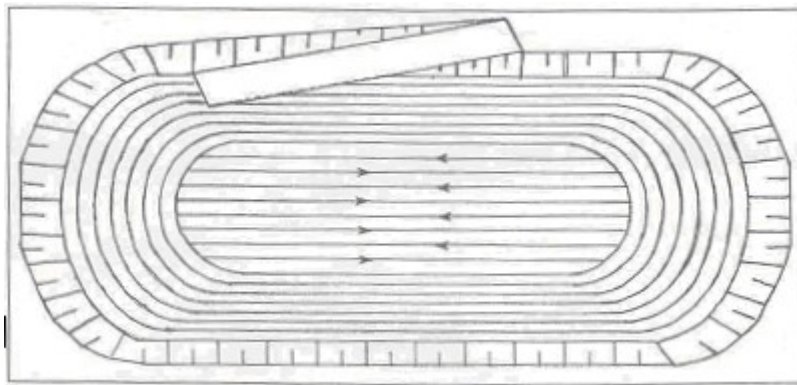
- Available length of the bench is more than 200m.
- Turning time is less than the empty travel time for the machine.
- Required turning space is available at the cut ends.



Turn Back

5.6 Continuous Mining Method

Continuous mining method ensures continuous cutting operation by the machine. In this method, the machine moves with cutting the material and near the pit ends it takes turn with a gentle angle while still cutting the material. Thus there is no discontinuity in cutting operation. This results in a round shape of the cut area particularly at the cut ends. The process continues till the circular/elliptical turnings become so sharp that the machine cannot take turn while continuing the cutting operation. Then the machine follows the turn back method for mining the rest of the material at the central portion. Overlapping circular or elliptical movement of the machine may be adopted to avoid this situation, but this creates problem like reduced productivity, difficulty in controlling the pit slope and floor gradient.



Continuous mining method

There are basically two modes of operation possible by Wirtgen Surface Miner on the basis of loading it can adopt – **conveyor loading and windrowing**.

In conveyor loading mode, the material that is cut by the machine is discharged by the discharge conveyor, which may load shiftable bench conveyor or large capacity off-highway trucks, or side-cast it. On the other hand, in windrowing mode, the machine cuts the material and the cut material is allowed to heap behind the machine in form of windrows. The cut material from the windrow or the side-casted heap can be loaded later conveniently to dumper/tipper by some loading equipment like FEL.

6.0 PRODUCTION CAPACITY OF SURFACE MINER

The production capacity of surface miners in different modes of working is calculated as shown in Table below.

Table - Production capacity of surface miner(after Dey, 1999)

Method of working	Windrowing Mode	Conveyor Loading Mode
Empty Travel Back	$\text{Planned Production (m}^3\text{)} = \frac{S \times L \times d \left(\frac{W \times 60}{L/v + L/v_e} \right)}{1000}$	$\text{Planned Production (m}^3\text{)} = \frac{S \times d \left(\frac{W \times 60}{L/v + t_c/L_t + t_e/L} \right)}{1000}$

Method of working	Windrowing Mode	Conveyor Loading Mode
Turn Back	$\text{Planned production (m}^3\text{)} = \frac{S \times L \times d \left(\frac{W \times 60}{\frac{L}{v} + tt} \right)}{1000}$	$\text{Planned production (m}^3\text{)} = \frac{S \times d \left(\frac{W \times 60}{\frac{1}{v} + \frac{tc}{Lt} + \frac{tt}{L}} \right)}{1000}$
Continuous	$\text{Planned production (m}^3\text{)} = \frac{S \times v \times d \times W \times 60}{1000}$	$\text{Planned production (m}^3\text{)} = \frac{S \times d \left(\frac{W \times 60}{\frac{1}{v} + \frac{tc}{Lt}} \right)}{1000}$
Where, L = Length of the face (m) D = Predetermined depth of cut (mm) v _e = Empty travel speed (m/min) W = Working hour available in a shift (hr) L _t = Length of cut to fill one truck (m) = (truck capacity in cu.m. x fill factor)/(S x d x swell factor/1000)		S = Width of the cutting drum (m) v = Machine speed during cutting (m/min) tc = Truck changing time (min) t _e = Empty travel back time (= L/v _e) (min.) tt = Machine turning time (min)

6.1 Geometrical Factors (Length of working area)

The productivity of a surface miner depends on the length of the available working area. Longer cuts will enhance the productivity, because only a smaller amount of the time is spent in maneuvering from one cut to the next. The forward speed (and therefore travel time for a given cutting length) furthermore depends on a number of factors:

- Cutting depth
- Material hardness and structure
- Type of machine and installed engine power

Therefore, the graph as shown in Fig.below only can show the general relation between effective production and cutting field length.

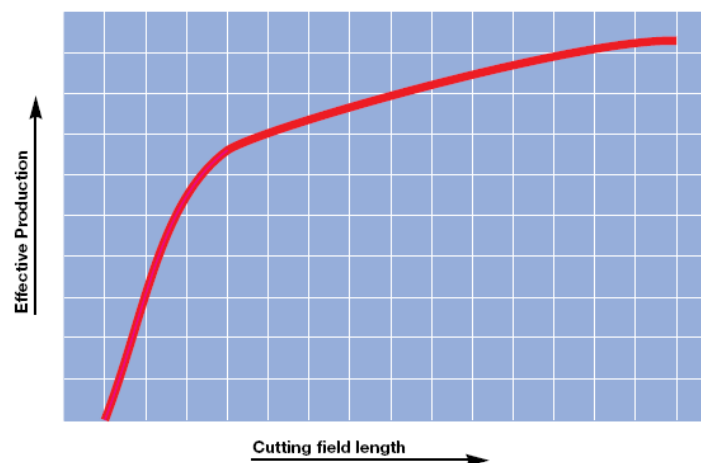


Fig.- Cutting field length vs. effective production

In standard applications, the appropriate minimum cutting field length should be in the range of:

- 100 m (hard material, low forward speed)
- 300 m (softer material, high forward speed)

Production range (cutting performance) of the surface miner 2200 SM in different modes of cutting is shown in Fig. below.

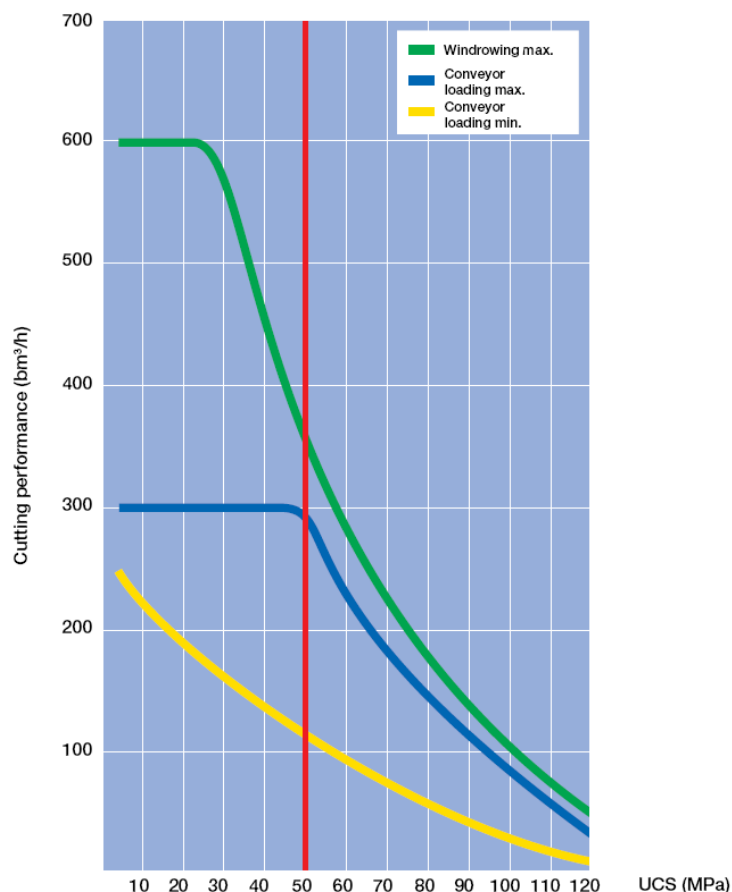


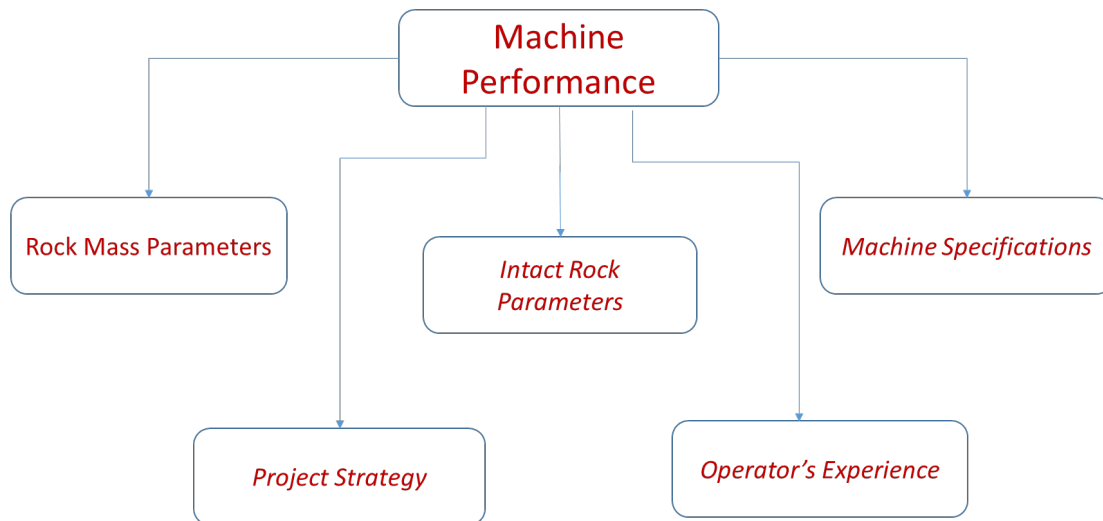
Fig.- Material with hardness > 50 MPa can only be cut under certain conditions.

7.0 PREDICTING SURFACE MINER PRODUCTION IN INDIAN CONDITIONS

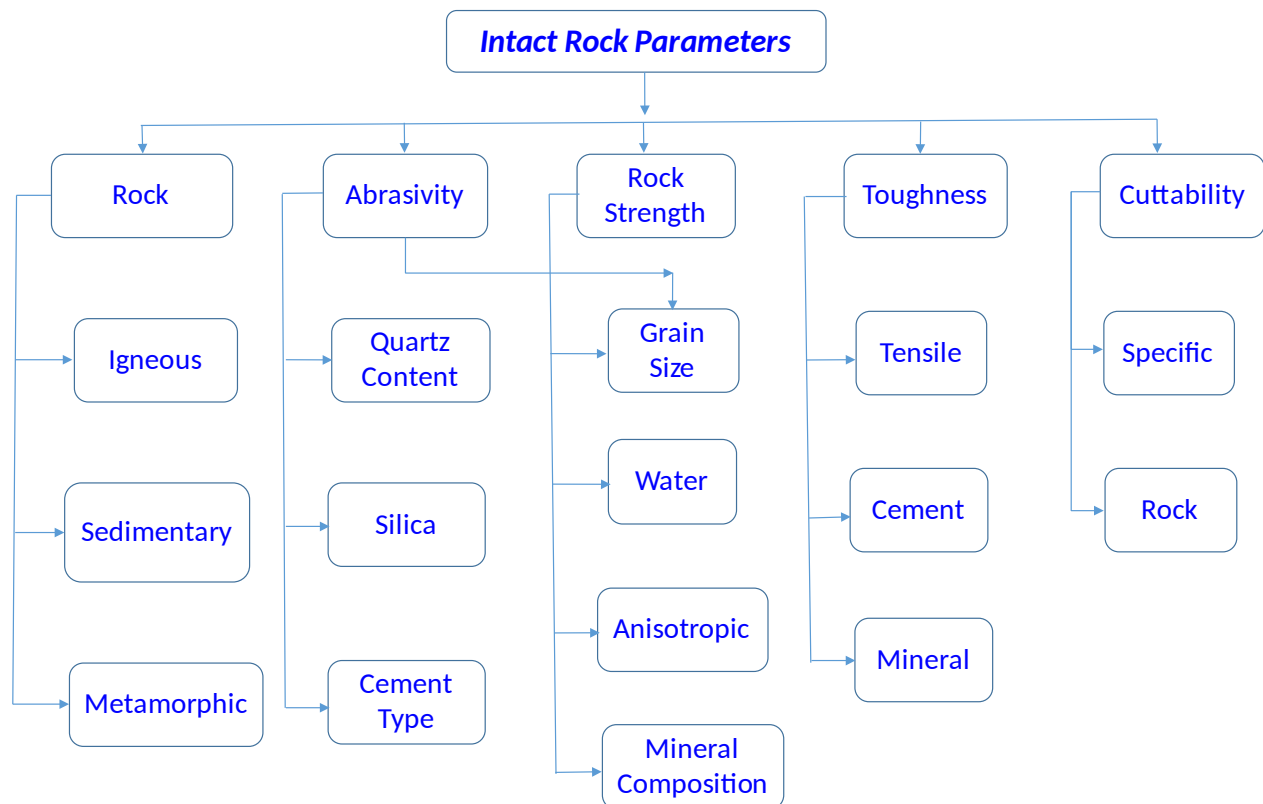
Successful application of surface miner technology largely depends on the selection of the most suitable equipment and cutting tools for the rocks and ground conditions to be encountered. Thus, rock properties assume significant importance.

7.1 Factors Influencing Surface Miner Performance

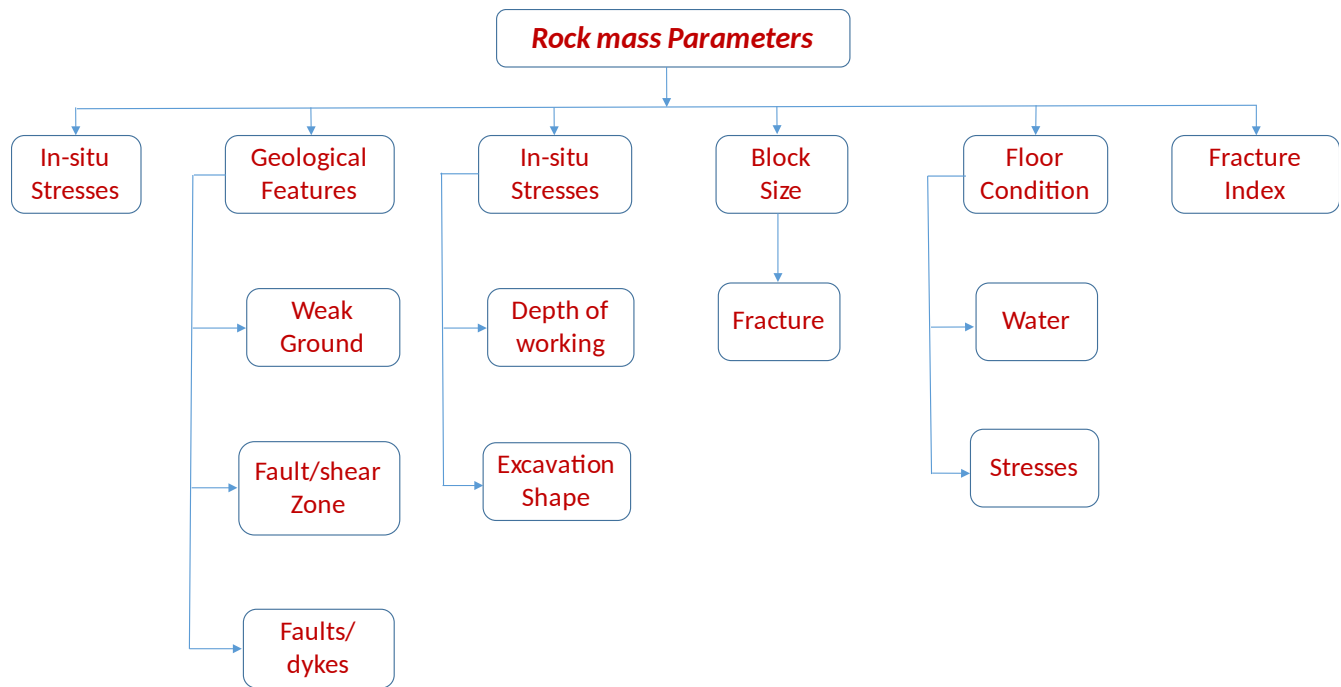
Numerous factors influence surface miner performance. Some of the main factors influencing the surface miner performance are given in Figures below



Parameters influencing machine performance



Intact rock parameters influencing machine performance



Rock mass parameters influencing machine performance

7.2 Rock Properties and Their Influence

The performance of surface miner mainly depends on,

1. Rockmass properties: rock strength and abrasivity, inclination and orientation of geological discontinuities etc.
2. Machine parameters: design of cutting head, type of cutters, machine power etc.
3. Mode of cutting: experience, job organization, machine facilities etc.

The following sections describe the influence of various rock/rockmass parameters on performance of surface miner from previously established relationships (Kramadibrata, Jones, 1993).

7.3 Uniaxial Compressive Strength (UCS)

The variation in cutting performance with uniaxial compressive strength is given in Fig. below.

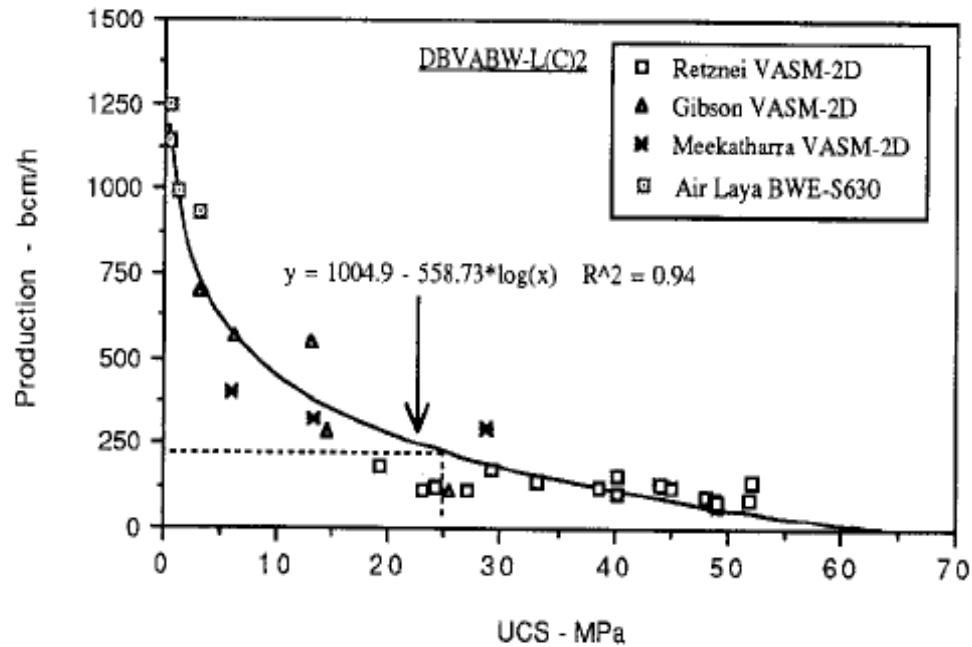


Fig. – Effect of UCS on cutting performance of Surface Miner

7.4 Brazilian tensile strength (BTS)

Rock being weak in tension most of the cutting systems utilize this weakness for failing the rock in tensile mode. Thus, tensile strength is an important parameter and is determined by Brazilian tensile strength. It's influence on production is shown in Fig. below.

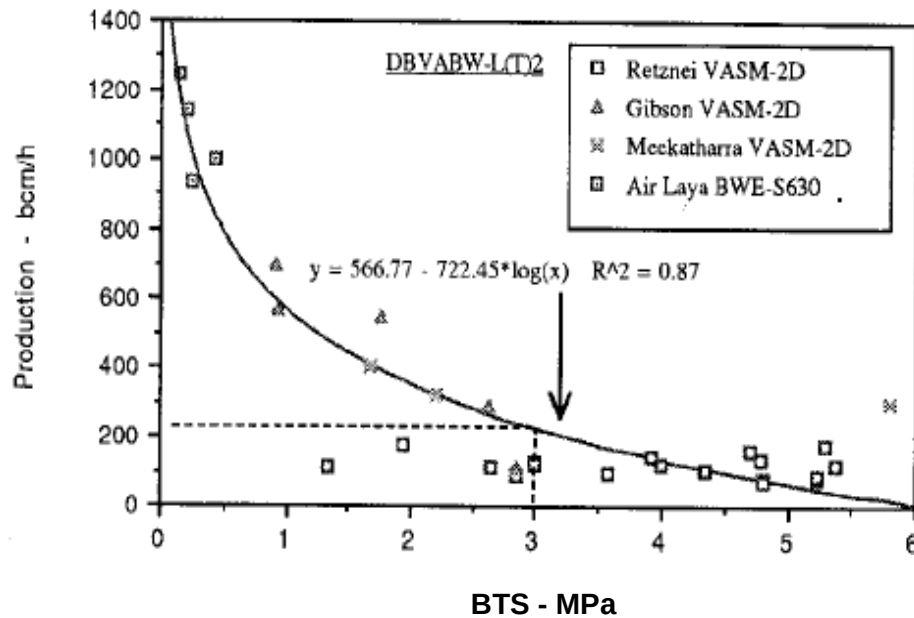


Fig. – Effect of BTS on cutting performance of Surface Miner

7.5 Point Load Strength Index (PLSI)

Point load strength closely resembles the cutting action of point attack picks mounted on the drum of surface miner. Its influence is shown in Fig.below.

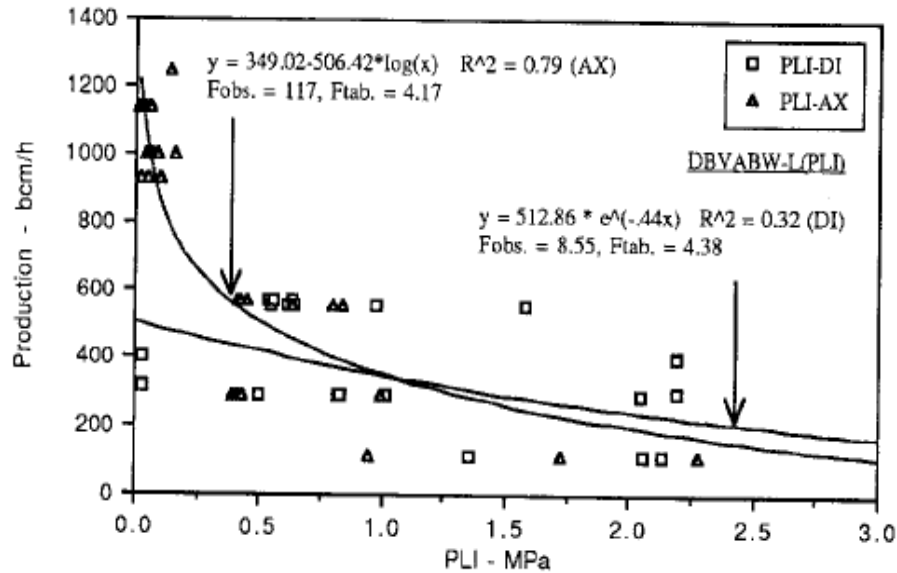


Fig. – Effect of PLSI on cutting performance of Surface Miner

7.6 Modulus of Elasticity (MOE)

Modulus of elasticity describes the strain induced in a rock sample for a given stress. For rock cutting system design this is a vital information. This is best obtained by young's modulus of rocks. Influence of Young's modulus is also well established as shown in Fig.below.

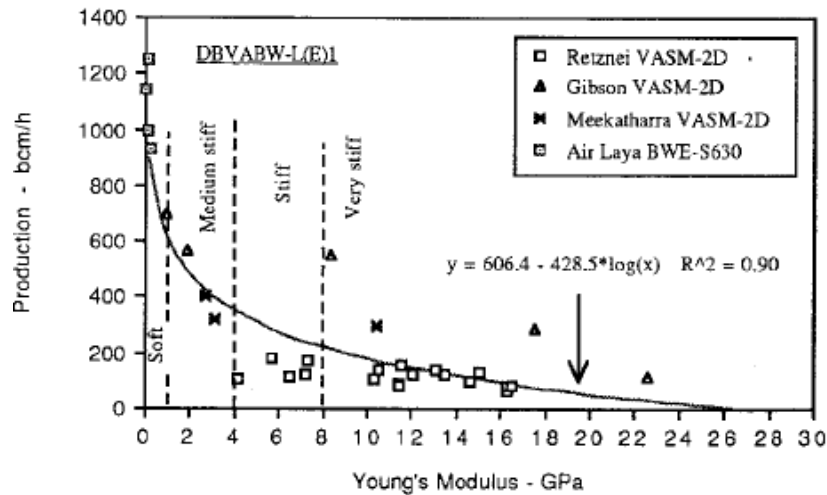


Fig.– Effect of MOE on cutting performance of Surface Miner

7.7 Sonic Velocity (SV)

Sonic velocity of rocks describes the degree of intactness/presence of minor defects at grain level. It's influence on production is shown in Fig.below.

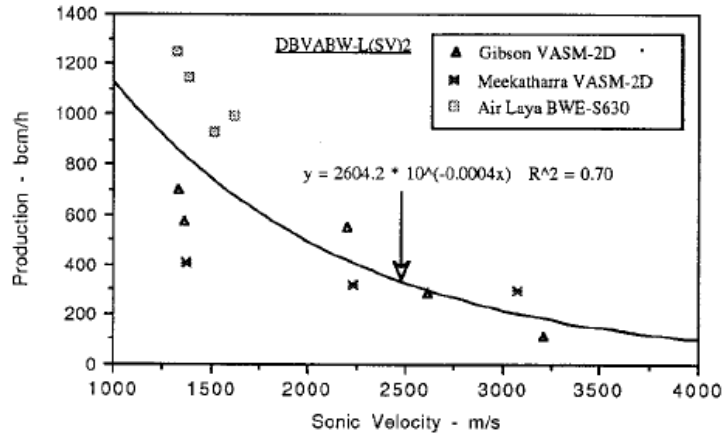


Fig. – Effect of SV on cutting performance of Surface Miner

7.8 Toughness Index (TI):

Toughness index is a measure of elastic strain energy involved in deforming rock with a cutting tool. The toughness(TI) is defined as:

$$TI = (\sigma_c^2 / 2E) * 100$$

Where,

TI= Toughness Index, MPa; σ_c = UCS, MPa, E = Young's Modulus, MPa

Higher the toughness more is the energy required to break the rock and less will be the production (Fig.below).

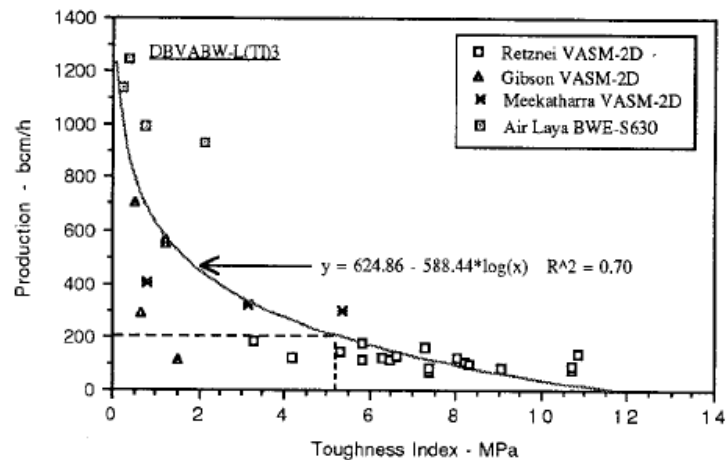


Fig. – Effect of TI on cutting performance of Surface Miner

Rock toughness index values less than 1 may indicate brittle, easily broken rocks. Values more than 2.5 indicate tough, difficult to cut rocks.

7.9 Brittleness Index (BI)

Brittleness Index is defined as the ratio of UCS and BTS. Rock has been classified on the basis of brittleness index as given in Table below. Tough rocks are easier to cut than brittle rocks. Effect of brittleness index on production is shown in Fig. below.

Table – Brittleness Index Classification (after Gehring, 1992)

BI	Description
6-7	Very tough and plastic
7-8	Tough and plastic
8-12	Average
12-15	Very brittle not plastic
15-20	Very Brittle

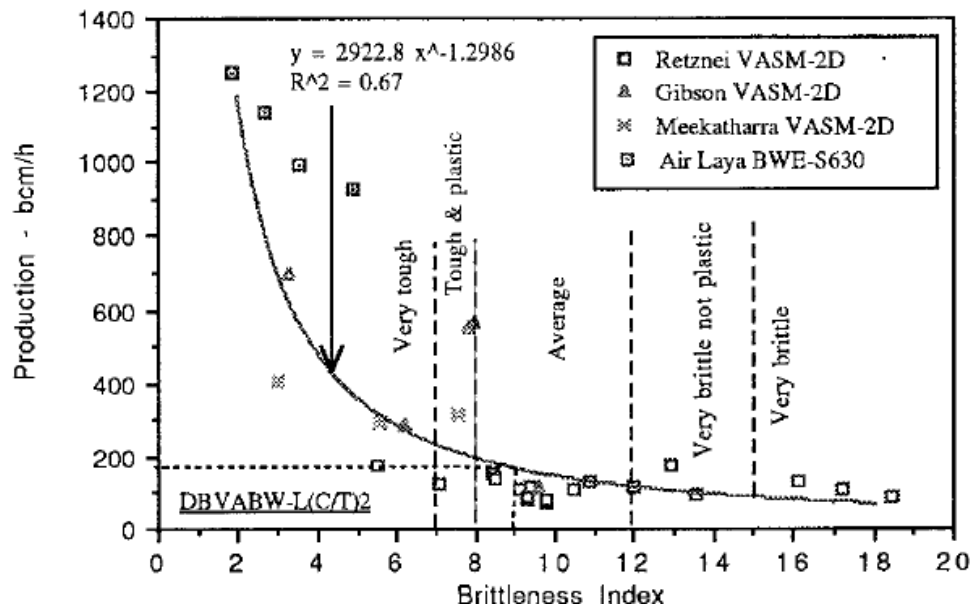


Fig. – Effect of BI Index on cutting performance of Surface Miner

7.10 Specific Fracture Energy (SFE)

The specific fracture energy required to destruct a rock sample is determined as the area under the stress-strain curve, and for the coal samples tested from LOCP, with average compressive strength of 20 MPa and average modulus of elasticity of 4.6 GPa this values comes to be 0.043 MJ/cu.m. For this energy the performance of the cutting system would be 230 cu.m./hour using a surface miner(VASM-2D) (Fig.below). However, in practice this has to be read alongwith RQD, Joint Spacing, rock toughness and silica content of rock apart from machine parameters.

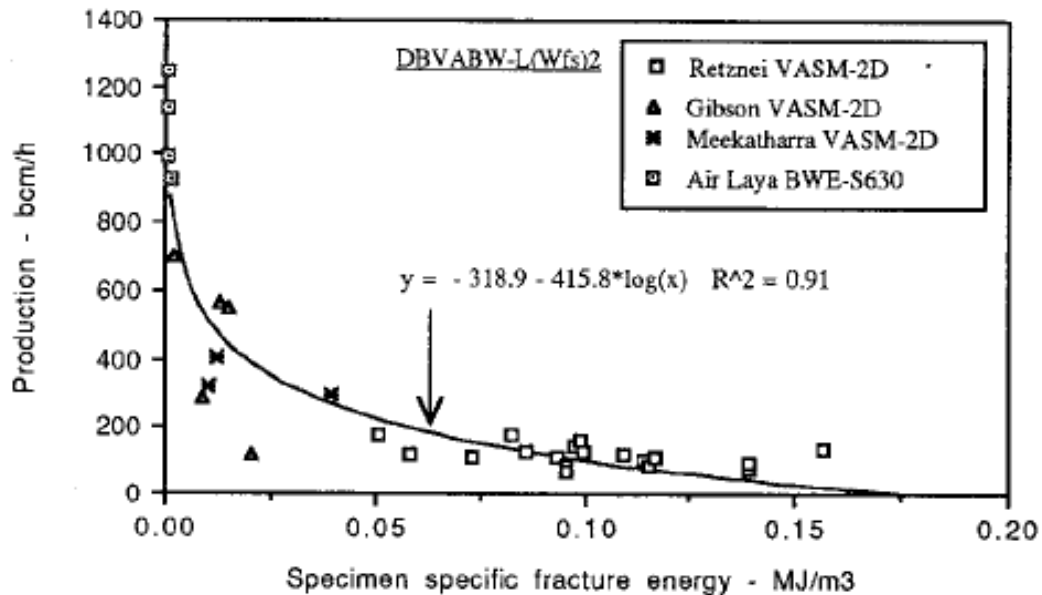


Fig. – Influence of specific fracture energy on cutting performance

The governing relationships, as established from previous research works, have been summarized in Table below.

Table – Regression equations relating production rates of surface miners and rock properties (Kramadibrata and Jones, 1993)

No.	Independent variables	Regression equations	r^2
1.	UCS (σ_c)	$L^* = 1004.9 - 558.73 \log (\sigma_c)$	0.94
2.	BTS (σ_t)	$L^* = 566.8 - 722.45 \log ((\sigma_t)$	0.87
3.	Apparent discontinuity spacing (Jap)	$L^* = 543 (Jap)^{-0.84}$	0.76
4.	Brittleness Index (BI)	$L^* = 2292 (BI)^{-1.2986}$	0.67
5.	[1/ Brittleness Index]	$L^* = -91.84 + 2657 (1/BI)$	0.73
6.	Young's Modulus (E_y)	$L^* = 606.4 - 428.53 \log E_y$	0.90

7.	Toughness Index (TI)	$L^* = 624.86 - 588.4 (TI)$	0.70
8.	Sonic Velocity (S_v)	$L^* = 2604 \cdot 10^{-0.0004(S_v)}$	0.70

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